1. Research question and principal findings

This experiment separates running an equity-report system from establishing that its output is reliable. Seven distinct companies have real eight-agent baseline and ninth-stage verifier outputs; five constitute the formal semiconductor cohort. NVIDIA has two additional independent baseline generations, making nine actual baseline runs. All consume saved real observations. Frozen-input reruns, saved-response revalidation and actual new generations are labeled separately.

Twenty selected financial observations for NVIDIA and AMD reconcile to accession-specific SEC filings: sixteen match directly and four EBITDA-related differences are explained by explicit definition bridges. Independently implemented EV/EBITDA and DCF arithmetic reproduces 84 engine components across seven companies. Neither result endorses the forecast: the default EBITDA formula omits R&D, historical multiple valuation is undiscounted, and debt is assumed to be 10% of estimated enterprise value.

The original V6 evaluation contains 1,478 baseline atomic assertions across all 24 agent sections. Only 291 (19.69%) are supported by the actual generating prompts; 40 contradict them. The verifier-treated versions contain 1,590 assertions, including 358 supported and 48 contradicted. Its useful measured contribution is transparent warning coverage: 93.99% of final Unsupported/Contradicted assertions receive an unverified warning, with 78.78% precision. Seventy-four problematic assertions remain unflagged, including eight contradictions. This is an improvement in issue disclosure, not a demonstrated reliable correction system.

The latest three-company follow-up is separately measured in Sections 9.2-9.3: after explicit evaluator corrections, grounding changes from 19.34% to 28.72%, with 31 versus 0 contradicted claims. Unsupported claims and machine-evaluation limitations remain. Earlier V6/V7.1 tables below retain their original evaluation version and must not be mixed with corrected denominators.

2. Implementation, environment and evidence boundary

The course supplied wrappers and audit headers rather than the full implementation. The actual upstream data processor, eight typed agents, valuation, sensitivity and rendering code were inspected at the fixed commit. The older AutoGen annual-report tutorial was not substituted. The upstream CLI also contains a plain-chat path; `experiment.py` deliberately invokes the actual eight typed agent implementations described by the course.

The dedicated Windows runtime is CPython 3.11.15; the complete recorded dependency set is `requirements-lock.txt` and environment details are in `evidence/preflight_configured/preflight.json`. Writer model: OpenAI `gpt-4.1-mini-2025-04-14`, temperature 0.7, 4,000 maximum output tokens per agent, structured outputs, SDK retries disabled. The original V6 verifier uses `gpt-4.1-2025-04-14`, temperature 0, up to 26,000 output tokens and at most two explicit validation repairs. Exact per-attempt settings, prompts, instructions, schemas, wire responses, usage, timings and failures are saved. Private configuration stays outside the course directory.

Compatibility repairs are separate from the single research improvement, Option A. They adapt entitled FMP stable endpoints, roll forecast labels forward from the last actual, expose missing inputs and engine assumptions, and render exact saved text with frozen market and valuation fields. AVGO/QCOM also need a public-source adapter after real FMP 402 denials. These are shared execution/data-availability changes, not competing verifier treatments. Public-source provenance is appended to those two writer prompts and saved; the three-company comparison uses unchanged original FMP-based prompts.

The final measurement uses explicit Codex-assisted claim adjudication under a saved common protocol. The task-agent model ID, token use and runtime are not exported, so they are unavailable, not zero. Earlier GPT-4.1 API judge proposals had extraction/semantic defects and remain as development evidence with costs. The machine adjudicator saw each complete source unit and its original prompt, decomposed factual premises, and recorded reasons; it did not use verifier verdicts as labels. Exact unchanged after sentences inherit their previously reviewed facts; all changed sentences have an explicit separate review. Shared model-family biases and development on the evaluated cohort limit independence. No human review took place.

The render manifest binds every final HTML section to the audited text hash. Later filings or online facts cannot make an original unsupported assertion supported. Offline replay rejects network connections and absent cached

requests; it is neither a fresh observation nor another independent generation.

Field	Source and interpretation
Cover price	Frozen FMP observation, or public Yahoo daily close for AVGO/QCOM
Cover target	FMP consensus target, not an LLM or engine target; missing for public adapter names
Cover grade	FMP analyst grade/action; neither LLM recommendation nor financial-health score
Engine target	Explicit EV/EBITDA, peer or DCF method with separate assumptions
Narrative	Eight structured writer outputs; ninth-stage revision separately mapped and audited
Public-source EBITDA	GAAP operating income + cash-flow D&A; differs from FMP peer definition

Legacy names also matter: an upstream `fwd_pe` field holds a historical ratio, and a net-debt/equity label can contain debt/equity. Renderer labels disclose these distinctions. `docs/field_lineage.csv` and `docs/data_valuation_source_audit.md` give source locations and formulas.

3. Industry universe, peers and actual coverage

The formal universe is semiconductor products, defined by material processor, accelerator, networking or wireless integrated-circuit businesses. SEC SIC is 3674 for NVDA/AMD/INTC/AVGO and 3663 for QCOM. Qualcomm's QCT platform segment is $38.367 billion of $44.284 billion FY2025 revenue (86.64%); Broadcom's semiconductor segment is $36.858 billion of $63.887 billion (57.69%). These support business-based membership while licensing and software constrain comparability. Sources: Qualcomm FY2025 10-K and Broadcom FY2025 10-K.

Company	Actual peers	Common-window price return	Important comparability limit
NVDA	AMD, INTC	+25.18%	GPU/CPU/system mix and software ecosystem differ across peers.
AMD	NVDA, INTC	+182.97%	Acquisition amortization and embedded exposure affect consolidated margins.
INTC	NVDA, AMD	+187.87%	Foundry losses/capital spending and intersegment eliminations differ from fabless peers.
AVGO	NVDA, AMD	+11.37%	Infrastructure software is 42% of FY2025 revenue; consolidated multiples are not pure semiconductor comparables.
QCOM	AMD, INTC	+14.27%	SEC SIC is 3663 rather than 3674; QTL licensing and handset mix limit consolidated comparability.

The common window is October 10, 2025 to October 9, 2026. All five raw-close returns are positive; their magnitudes differ, but this is not balanced winner/loser coverage. Dividends are excluded and provider split conventions are not independently verified. FMP supplies the first three series and public Yahoo chart responses the last two. AAPL and GOOGL are onboarding-only, with cross-industry demonstration peers, and never count toward the formal five.

FMP returned 402 for AVGO/QCOM financials and for news access. Existing WRDS web exports were privately retained, but the user chose public materials after review of WRDS AI restrictions; no WRDS observations enter a model or this submission. Public AVGO/QCOM inputs use exact SEC 10-K comparative FY2023-2025 facts, matching iXBRL contexts, dated cover-page shares, and a saved Yahoo price response. AVGO D&A includes right-of-use amortization. Missing subject multiples remain missing, and the valuation fallback is explicitly 12x with a 3x spread, not a measured multiple. Their FMP peer snapshots are real and frozen but have different EBITDA definitions and fiscal calendars.

All five run sensitivity, catalyst and enhanced-news functions. News substitutes are narrow: one actual issuer release for the original FMP companies, one literal current SEC 10-Q income-statement table for AVGO/QCOM. They are not comprehensive press coverage. The latter table flattening creates quarter-versus-year-to-date ambiguity that the writer sometimes misreads. Native default earnings dates, heuristic 95% event probabilities

and assumed 15% scenario standard deviations are explicitly assumptions. The original INTC/AAPL news-filter omission is preserved alongside the repaired `news_v2` analyses. The public data cutoff is an analytical clock, while HTTP fetches occurred later that day; this is a contemporary snapshot exercise, not proof of historical point-in-time availability or a backtest.

4. Data audit: twenty observations with original SEC contexts

The audit uses NVIDIA FY2026, January 27, 2025-January 25, 2026, accession `0001045810-26-000021`, filed February 25, 2026; and AMD FY2025, December 29, 2024-December 27, 2025, accession `0000002488-26-000018`, filed February 4, 2026. Fiscal-year labels, observation dates and publication dates are separate. A SEC companyfacts calendar frame is not a replacement for the issuer's fiscal year. The original filings and corresponding API facts were saved with timestamps and SHA-256 hashes. Each source ingredient is linked to an exact JSON pointer and a matching dimension-free iXBRL fact/context in the original filing.

For each company the selected pipeline metrics are revenue, cost of operations, SG&A, contribution profit, contribution margin, EBITDA, EBITDA margin, SG&A margin, revenue growth and basic EPS. The contribution label represents gross profit under the upstream equation, not a variable-cost contribution measure. Ratios are independently calculated from filing ingredients; market multiples are excluded from this SEC-only ten-item list because they additionally require a dated market price and enterprise-value inputs. The full twenty-row `audits/data/data_audit.csv` includes values, differences, units, periods, verdicts, formulas, accessions, links and contexts.

The observed result is sixteen direct matches within display precision and four reconciled definition differences. Margins have a 0.05-percentage-point tolerance because the pipeline displays one decimal; basic EPS has a $0.005 tolerance; USD totals have a $0.50 tolerance. None of the selected twenty observations remains unexplained.

EBITDA bridge, $million	**NVDA FY2026**	**AMD FY2025**
GAAP operating income	130,387	3,694
D&A	2,843	3,004
Operating-income-plus-D&A reference	133,230	6,698
Net income	120,067	4,335
Signed income-tax expense/(benefit)	21,383	-103
Interest expense	259	131
NI + tax + interest + D&A	144,552	7,367
Discontinued-operations income excluded in AMD bridge	–	66
Equity-method income excluded in AMD bridge	–	26
Actual pipeline FMP EBITDA	144,552	7,275
Pipeline minus operating reference	11,322	577
Unexplained residual after identified formula	0	0

NVIDIA's FMP value is exactly reproduced by net income plus tax, interest expense and D&A. The $11.322 billion difference from operating EBITDA therefore reflects non-operating earnings other than interest expense. AMD requires a different bridge: the simple net-income formula is $92 million too high; excluding $66 million discontinued-operations income and $26 million equity income gives the observed $7.275 billion. Equivalently, $4.140 billion continuing pretax income before equity income plus $0.131 billion interest plus $3.004 billion D&A equals $7.275 billion. The intermediate EBIT of $4.271 billion also matches FMP. Sources: NVIDIA consolidated statements and cash flows and AMD consolidated statements and cash flows.

This empirical reconciliation should not be generalized into a universal provider EBITDA policy. It shows why a standardized field name can still obscure economically different non-operating exclusions across issuers. For AMD, D&A includes $750 million other D&A and $2.254 billion acquisition amortization. FMP's $17.487 billion cost of revenue also equals the printed $16.456 billion cost-of-sales line plus $1.031 billion cost-of-sales amortization. Selecting only the smaller printed line would create a false data-error finding.

The AMD 10-K/A corrects transposed MD&A shipment and selling-price growth percentages, not the audited financial statements used here. Both versions are retained, so textual use of that MD&A passage must use the amendment. No later source was injected into baseline prompts to retroactively improve grounding classifications.

5. Forecast audit: a definition break larger than the stated margin assumption

The original code fixes forecast labels at 2025E/2026E/2027E. With newer actuals this overlaps history. The compatibility mapping preserves the original settings but assigns the next three years after the latest actual: NVDA 2027E-2029E and AMD 2026E-2028E. It does not change the default growth path of 5%, 6%, 4%, which compounds to 1.15752 times starting revenue over three years.

Contribution margin increases by one percentage point annually and SG&A/revenue decreases by half a percentage point annually, starting from rounded historical ratios. These are absolute percentage-point movements, not relative percentages. A later calculation overwrites projected EBITDA with contribution profit minus SG&A, excluding R&D and other expenses. The supplied EBITDA-margin increment affects an EPS shortcut but does not control the final EBITDA series as a reader might expect.

Company / fiscal year	Revenue, $bn	Gross/contribution margin	SG&A/revenue	Final EBITDA margin
NVDA 2027E	226.735	72.1%	1.6%	70.5%
NVDA 2028E	240.339	73.1%	1.1%	72.0%
NVDA 2029E	249.953	74.1%	0.6%	73.5%
AMD 2026E	36.371	50.5%	11.5%	39.0%
AMD 2027E	38.553	51.5%	11.0%	40.5%
AMD 2028E	40.095	52.5%	10.5%	42.0%

AMD's actual FMP EBITDA margin is 21.002%; its operating-income-plus-D&A margin is 19.337%. Neither definition supports treating the jump to 39.0% as a routine one-point improvement. NVIDIA's SG&A/revenue declines from 2.1% to 0.6%; the code supplies no company-specific evidence that this expense compression is attainable. Historical revenue growth was 65.47% for NVIDIA and 34.34% for AMD, substantially different from the common immediate 5% path. Those observations motivate questioning the assumptions, not extrapolating past growth indefinitely.

The analyst-selected alternative growth paths and unchanged initial expense ratios are now executed in a separate frozen-input parameter exercise in Section 5.1. Their source-based rationale is retained, but neither scenario is company guidance or a fitted forecast. The original Option A experiment continues to use its unchanged baseline assumptions.

A second measured inconsistency is in the separate peer projection: it averages historical annual EBITDA growth and applies that rate once, labeling every company with the latest actual year across the table plus one. In NVDA's prompt, its own 2027 peer-table EBITDA is $382.001 billion while the main 2027 forecast is $159.848 billion, a 138.98% difference. AMD and INTC are labeled 2027 from 2025 actuals after only one growth step. `audits/peer_projection_audit.csv` independently reproduces all fifteen peer projections in the five available snapshots. A statement can match a cell and still conceal contradictory assumptions elsewhere in the same evidence.

The cutoff is a current-data date. FY2027 for NVIDIA and FY2026 for AMD are already partially elapsed, and quarterly releases exist. This is not an out-of-sample historical forecasting experiment; using subsequently known results to justify an older forecast would be invalid.

5.1 Executed offline parameter scenarios

To test the proposed assumptions, I separately invoked the same pinned forecasting functions on frozen NVIDIA and AMD raw statements. Each company has a default replay and an analyst-deceleration case. NVIDIA uses sequential annual revenue growth of 30%, 20%, 10%; AMD uses 15%, 10%, 8%. Both annual contribution/EBITDA-margin input increments and the SG&A-margin change are set to zero. The adjusted contribution and SG&A ratios therefore stay at the processor's rounded latest-actual levels: 71.1% and 2.1% for NVIDIA, 49.5% and 12.0% for AMD. No precise SEC ratio was silently substituted for a rounded pipeline input.

Company / fiscal forecast	Default revenue, $bn	Adjusted revenue, $bn	Default EBITDA margin	Adjusted EBITDA margin
NVDA 2027E	226.735	280.719	70.5%	69.0%
NVDA 2028E	240.339	336.863	72.0%	69.0%
NVDA 2029E	249.953	370.550	73.5%	69.0%
AMD 2026E	36.371	39.835	39.0%	37.5%
AMD 2027E	38.553	43.818	40.5%	37.5%
AMD 2028E	40.095	47.324	42.0%	37.5%

The rationale remains a deceleration from observed annual growth of 65.47% and 34.34%, without assuming the same immediate 5% rate for both businesses or further expense-ratio compression. These are analyst-selected conditional assumptions, not guidance, fitted predictions or validated forecast accuracy. All six forecast periods remain labeled as issuer fiscal years.

The same EV/EBITDA engine uses third-year EBITDA, the original historical mean multiple, original shares and the original rule that assumed net debt equals 10% of each scenario's EV. Debt amounts consequently change; the rule is held fixed, not the dollar debt. Neither discounting nor terminal growth enters this multiple method.

Company / EBITDA year	Default model target, USD	Adjusted model target, USD	Change
NVDA 2029E	333.71	464.43	+39.17%
AMD 2028E	377.84	398.17	+5.38%

NVIDIA's terminal revenue increases 48.25% while its terminal EBITDA margin falls from 73.5% to 69.0%; AMD's revenue increases 18.03% while margin falls from 42.0% to 37.5%. With shares and the multiple unchanged, the target ratio equals the terminal revenue ratio times the EBITDA-margin ratio. Thus stronger growth and lower margins jointly produce the displayed target changes. Those prices are conditional model outputs, not endorsed fair values. Zero margin improvement does not repair the upstream definition: projected EBITDA still omits R&D and does not explicitly reconcile D&A, and AMD's adjusted 37.5% remains far above its 21.002% historical provider measure.

The default replays match all 220 saved table fields. The scenario implementation records 140 internal arithmetic checks; a separately written standard-library Decimal reference passes 670 additional numerical and integrity checks. Full configurations, actual processor CSVs, engine output, annual deltas, assumptions, source hashes and logs are in `audits/forecast_parameter_scenarios/run_v2/`. The run took 1.84 seconds including integrity hashing, with zero API calls, tokens or new API cost. Python network/subprocess rejection probes passed; this is not an OS sandbox. A second check confirms all 9,083 protected original submission files are byte-identical, including snapshots, writer prompts, generated texts, verifier treatments, paired claim audits and the API ledger. This fulfills the forecast-audit parameter exercise and does not introduce a second research improvement or modify the Option A comparison.

6. Independent valuation and genuine target-price sensitivity

The independent notebook imports only the submitted arithmetic script, not `ValuationEngine`. EV/EBITDA is reconstructed as:

`EV = terminal-forecast EBITDA × historical mean EV/EBITDA`
`assumed net debt = 10% × EV`
`equity target per share = max(0.9 × EV / observed shares, 0).`

The multiple is an unweighted historical own-company mean, not an independently selected fair peer multiple. The third-horizon EBITDA is not discounted. The debt term is assumed from value, rather than measured from debt less cash. Low/high estimates vary the multiple by its historical sample standard deviation while keeping central debt fixed. These are mechanics to reproduce, not assumptions endorsed by this study.

The independent DCF uses supplied latest-annual FCF: $96.676 billion for NVIDIA, $6.735 billion for AMD, and negative $4.949 billion for Intel. Thus the engine's fallback of 60% of terminal forecast EBITDA exists in code but is not used in these three measured DCFs. The ten-year default grows FCF by 10% in years 1-5 and 5% in years 6-10, discounts at 10%, and uses 2.5% terminal growth. It again assumes net debt of 10% of modeled EV. Its reported low/high range changes only the terminal discount denominator by one percentage point; it is not a full WACC sensitivity. Intel's negative FCF propagates through the growth model and its negative equity estimate is clipped to a zero share target.

Company	Frozen quote, USD	EV/EBITDA target, USD	DCF target, USD	Forecast EBITDA period
NVDA	229.28	333.71	72.52	2029E
AMD	608.10	377.84	75.05	2028E
INTC	104.70	105.28	0.00	2028E
AAPL, onboarding only	336.64	306.15	122.96	2028E
GOOGL, onboarding only	351.66	319.75	110.00	2028E

All twelve compared values per company – target, low, high, EV, debt and equity for each of two methods – match the engine: eighty-four reconciled values across seven distinct companies (including the two onboarding-only names). The peer method is saved separately and is not falsely counted as an independent reproduction. Shares, FCF periods, multiples and all hidden assumptions are explicit in each `valuation_assumptions.csv`; projected DCF terms are in `dcf_cash_flows.csv`.

Nine saved notebooks, including the two compatibility-derived news snapshots, were also relocated to a minimal temporary submission and executed in an isolated Python process with network connections forbidden. They needed neither the cached upstream checkout nor pandas. All results matched within relative tolerance 1e-12 and absolute tolerance 1e-7; minor floating-point differences are recorded. Every original valuation CSV retained its pre-test hash. `evidence/notebook_portability_v6.json` and `scripts/check_notebook_portability.py` make this portability check repeatable.

The native sensitivity analyzer was run, but its combined grid is EBITDA, not a share target. We propagate its 25 combinations – revenue level changes from -5% to +5% and margin changes from -2 to +2 percentage points – through the same EV/EBITDA equations, holding the multiple and shares fixed and preserving the baseline's proportional-debt rule. The computation uses unrounded scenario EBITDA rather than the native display rounded to $0.1 billion.

Company	Base target	Target range, USD	Price change for +1 margin point	Revenue elasticity
NVDA	333.71	308.40-359.93	+1.36%	1.00
AMD	377.84	341.85-415.62	+2.38%	1.00
INTC	105.28	93.48-117.77	+3.27%	1.00

The revenue shock changes the forecast level, not an annual growth rate compounded through the forecast. A two-percentage-point margin shock means, for example, 42% to 44%, not 42% multiplied by 1.02. Away from zero clipping, price is proportional to revenue, so its local revenue elasticity is one; the impact of one margin point depends on the base margin. CSVs, native analyzer outputs and heatmaps are under `audits/target_price_sensitivity/` for NVDA/AMD/INTC and `audits/sensitivity/` for AVGO/QCOM.

These narrow scenario ranges are not confidence intervals. They omit uncertainty about the multiple, actual debt, forecast definition and timing. The difference between NVIDIA's $333.71 multiple target and $72.52 DCF target is much larger than its ±5% revenue/±2-point margin band. This is evidence that structural assumptions dominate the chosen local sensitivity exercise, not evidence favoring one of the two point estimates.

Additional publicly sourced semiconductor valuation results:

Company	Quote USD	EV/EBITDA target	DCF target	EBITDA horizon / multiple
AVGO	361.54	109.92	102.44	2028E; 12x fallback
QCOM	175.50	278.91	221.84	2028E; 12x fallback

All five formal target-price sensitivity results (relative revenue +/-5%, margin +/-2 percentage points):

Company	Base target USD	Target range USD	Price effect of +1pp margin
NVDA	333.71	308.40-359.93	1.36%
AMD	377.84	341.85-415.62	2.38%
INTC	105.28	93.48-117.77	3.27%
AVGO	109.92	101.25-118.93	1.52%
QCOM	278.91	254.95-303.93	1.89%

AVGO and QCOM shares are dated August 28 and July 27, 2026, respectively, rather than contemporaneous share counts. Their FCF is GAAP operating cash flow less capex, not demonstrated unlevered FCF. These additional assumptions limit economic interpretation of the reproduced targets.

7. Original V6 claim audit (historical evaluation)

The saved protocol covers all nonblank source units, with every unit receiving atomic factual premises or an explicit nonfactual reason. Baseline source-unit counts are 148 NVDA, 145 AMD and 146 INTC; after counts are 188, 175 and 158. This is complete textual coverage and explicit machine decomposition, not proof that every implicit semantic premise was recognized. Exact quotes, offsets, run IDs, original prompt hashes, evidence line ranges, classifications, numeric status and rationale are in the claim CSVs. A zero-claim section would be N/A; none occurs here.

Supported means entailed or arithmetically derivable from the original prompt. Unsupported means insufficient evidence, even if a later web search could establish truth. Contradicted requires an explicit conflict with supplied evidence. Numeric errors include wrong period, issuer, units and actual/forecast meaning, not just wrong digits. Qualitative causal and market-position claims require evidence too.

Company	Baseline S / total	Unsupported	Contradicted	Grounding
NVDA	81 / 486	388	17	16.67%
AMD	96 / 553	447	10	17.36%
INTC	114 / 439	312	13	25.97%
ALL	291 / 1,478	1,147	40	19.69%

Across all three companies, every section is included:

Section	Before S / total	Before U / C	After S / total	After U / C
tagline	14 / 28	13 / 1	16 / 30	14 / 0
company overview	28 / 295	261 / 6	26 / 281	249 / 6
investment overview	68 / 163	92 / 3	108 / 238	125 / 5
valuation overview	46 / 168	115 / 7	48 / 165	110 / 7
risks	20 / 250	226 / 4	22 / 249	223 / 4
competitor analysis	17 / 295	272 / 6	22 / 321	288 / 11
major takeaways	42 / 161	110 / 9	46 / 171	116 / 9
news summary	56 / 118	58 / 4	70 / 135	59 / 6

Competitor analysis has the weakest baseline support (17/295, 5.76%), followed by risks (20/250, 8.00%) and company overview (28/295, 9.49%). These agents request moats, management history, customer exposure and causal explanations from prompts primarily containing financial/peer tables and one issuer excerpt. The instructions invite evidence gaps. Investment update and news summary have relatively more supplied earnings facts, but still add consensus comparisons and causal claims not in those excerpts.

One concrete error is NVIDIA's historical contribution margin described as consistently above 70%, although its prompt includes 64.9% and 56.9%. Another repeatedly carries the historical 68.3% revenue CAGR forward through 2029, despite future annual assumptions of 5%, 6%, 4%. AMD's 20.5% actual CAGR likewise appears as if it describes the entire horizon through 2028. Exact table matching without period semantics does not detect these failures.

8. Three independent runs and stability

`NVDA_baseline_1`, `_2`, `_3` are three real generations: 24 paid writer calls with identical per-agent request hashes, frozen data, model and parameters, and distinct provider completion/call IDs. Risk extraction explicitly covers all 76 risk source units. It uses the first three distinct risk categories mentioned, with a fixed taxonomy and tie convention, not a claim of economic importance ranking.

Run	Explicit writer rating	Writer target	First three risk categories
1	Not stated	Not stated	Demand cyclicality; macro/currency; competition
2	Conditional hold or increase	Not stated	Demand cyclicality; revenue concentration; export controls/geopolitics
3	Overweight	Not stated	Demand cyclicality; revenue concentration; supply capacity

Writer target variance is N/A because no explicit per-share target is present. Rating variance is N/A because only one draw has a unique standard rating; one observation is not a zero-variance finding. The frozen FMP target is $339.17 in all three runs, with sample variance zero, which provides no evidence about LLM stability. Risk-set Jaccard overlaps are 0.20, 0.20 and 0.50. Kendall agreement over shared categories is N/A, N/A and 1.0; the last pair shares only two categories, so it is weak evidence. All quotes, risk mappings, missing fields and generation independence checks remain under `audits/stability/NVDA_adjudicated/`. Three draws cannot characterize the full output distribution.

9. Option A: original experiment and measured follow-ups

The ninth stage combines structured evidence review with deterministic citation and numeric-token binding. Code reconstructs verbatim prompt lines, rejects nonexistent or structural-only citations, checks complete source-unit/word coverage, and maps each original span to a retained, corrected, annotated or withheld release. Unsupported facts are visibly annotated, not approved on plausibility. A claim labeled Contradicted by the verifier must be corrected with evidence or withheld; a proposed correction that fails deterministic checks is not released. However, the verifier can itself misclassify a contradiction, which explains independently observed residual conflicts.

V1-V4 real failures exposed quote copying, number/period bindings, missing words and overly strict action validation. V5 allowed evidence-bounded rephrasing and revalidated saved model responses under immutable contracts. V6 rejects citations containing only a company identity, heading or table header. The first five V6 reports revalidate existing paid reviews without new model generation; AVGO/QCOM have new V6 calls. All earlier attempts and charges remain visible. This adaptive development uses the evaluated cohort and is not a preregistered held-out test. Source snapshots and version hashes preserve each iteration.

Before and after use the same original evidence and independent protocol for the complete eight sections of NVDA, AMD and INTC. The final HTML consumes exactly the after-audited text. The quantitative outcome is:

Company	Grounding before -> after	Contradiction before -> after	Verifiable numeric error before -> after
NVDA	16.67% -> 20.04%	3.50% -> 4.28%	18.97% -> 17.72%
AMD	17.36% -> 19.55%	1.81% -> 2.08%	9.26% -> 8.70%
INTC	25.97% -> 29.42%	2.96% -> 2.88%	15.25% -> 13.04%
ALL	19.69% -> 22.52%	2.71% -> 3.02%	14.62% -> 13.36%

Numeric errors are 25/171 verifiable numeric claims before and 29/217 after. Including unverifiable numeric claims in the denominator gives 25/205 (12.20%) and 29/258 (11.24%); both denominators are reported to prevent missing evidence from disguising error. Raw supported share rises 2.83 percentage points, but contradiction count rises from 40 to 48. More repeated assertions change the denominator: final text contains 155 duplicate lexical-key instances. These are not demonstrated unique-fact improvements.

Visible claim count is 107.58% of baseline; independently supported lexical fact-key retention is 96.13%, and supporting evidence-span retention is 100%. The verifier reports 99.80% visible retention but only 10.06% retained as verified. Lexical matching can miss paraphrases and table spans can support multiple facts, so neither proxy proves semantic fact recall. The claim mapping and remaining-error casebook must accompany these rates. The system did not improve grounding by deleting all information; it often preserves too much repeated unverified material.

Continued on next page

Company	Warning recall on U/C	Warning precision	Unflagged U/C	Unflagged C

Company	Warning recall on U/C	Warning precision	Unflagged U/C	Unflagged C
NVDA	94.40%	82.03%	23	7
AMD	94.62%	81.34%	27	0
INTC	92.48%	71.43%	24	1
ALL	93.99%	78.78%	74	8

Before treatment there are no warnings, so warning recall is zero and warning precision is N/A. After treatment, 1,158 of 1,232 problematic assertions are covered by warnings; 312 supported assertions are also inside warned paragraphs. Annotation coverage is a measurable review aid, but most text becomes warned and reviewer burden is high. Eight contradictions remain unflagged: examples include NVIDIA's latest comparable multiple being above AMD when the prompt shows 33.7774x versus 47.8456x, and actual 2025 values labeled as projections. Warning status is computed over the whole released paragraph, independently of verifier labels.

The appropriate conclusion is a partial engineering success and a semantic failure: the verifier exposes many evidence gaps while it remains unreliable as a correction gate. Residual errors and failed repairs are preserved, not cleaned out of the scorecard after seeing results. AVGO/QCOM and the two onboarding-only names have verifier outputs, but no full independent claim score is asserted for them. Contradicted content means the generated reports do not satisfy a strict zero-factual-error release standard. The research report describes those errors as experimental observations and does not recommend trading on them.

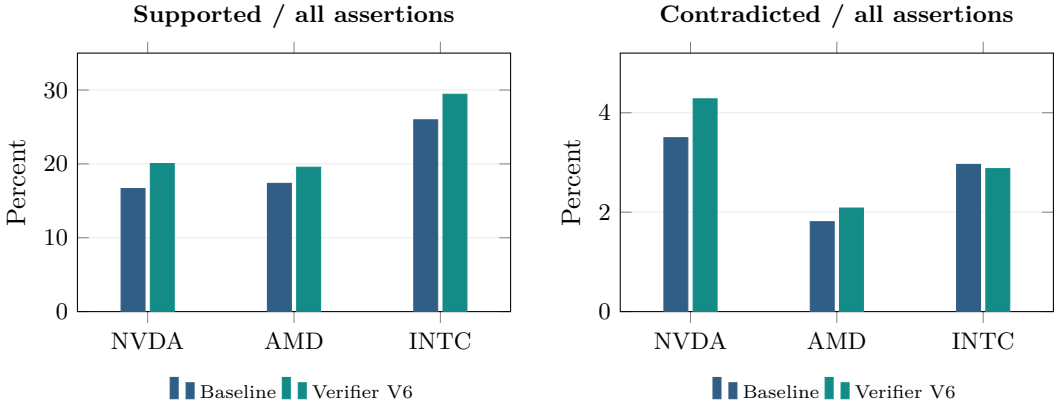


Figure 1: Measured grounding and contradiction rates on all 24 sections; repeated claims affect denominators. Historical V6 evaluation.

9.1 Development follow-up: improved warnings, unresolved semantics

The completed V7.1 follow-up is reported separately from the historical V6 release. V7 added semantic review instructions about periods, entities, comparisons, CAGR and universal trend claims, plus exact released-string deduplication. V7.1 repaired a bounded validation failure by permitting only three neutral connectors. Both changes remain within Option A. All seven companies received new V7 reviews; V7.1 reused proven saved responses and generated two missing Intel sections. Reuse is not independent generation. Development patches apply to separate clean pinned checkouts; they do not overwrite the main V6 patches.

The complete NVDA/AMD/INTC follow-up covers all eight sections with unchanged baseline evidence: 360 exactly unchanged sentence units carry independent labels with text/evidence identity, and all 173 changed units were explicitly decomposed and judged against the original writer prompts. These are machine-assisted labels, without human review. The task agent used the evaluated cohort to guide development; there is no held-out efficacy claim and no ablation separating prompt and deduplication effects.

Measure	Baseline	V6	V7.1 development
Supported / total	291 / 1478	358 / 1590	325 / 1673
Grounding rate	19.69%	22.52%	19.43%
Contradicted / total	40 / 1478	48 / 1590	41 / 1673
Contradiction rate	2.71%	3.02%	2.45%
Verifiable numeric errors	25 / 171	29 / 217	23 / 180
Warning recall	0%	93.99%	97.70%
Warning precision	N/A	78.78%	82.42%
Unflagged U/C	1187	74	31
Unflagged contradictions	40	8	8

V7.1 supported lexical-key retention is 76.06%, supporting evidence-span retention is 93.02%, and claim count is 113.19% of baseline. Verifiable numeric error rate is 12.78%; including unverifiable numeric claims gives 23/222 (10.36%). Wording, decomposition and duplicate removal change denominators; these are not unique semantic-fact recall estimates. Selected three-company review allocation is 186,938 tokens, $0.807346 and 276.42 seconds, including $0.059582 of new V7.1 calls. Global accounting also includes failed repairs and evaluation attempts. Full company/section tables, claim maps and retained failures are under `comparison_v7_1`, `annotation_v7_1` and `docs/verifier_v7_followup.md`.

The evaluator also failed. An unblinded GPT-4.1 proposal skipped 18 warned NVIDIA units as nonfactual. Hiding only the warning prefix eliminated those omissions but still under-decomposed claims and missed explicit evidence. Both proposal versions remain saved and neither supplies the canonical labels. Explicit task-agent judgments replaced them. Schema compliance and automated coverage alone do not establish valid financial labels.

A subsequent one-section NVIDIA pilot used pinned `gpt-5.4-2026-03-05`, high reasoning effort and a 26,000 completion-token limit, with identical V7.1 instructions and original frozen inputs. It returned HTTP 200 with `finish_reason=length`, 4,372 input tokens, 26,000 reasoning/output tokens and empty visible content. The SDK then raised MaxTurnsExceeded. No verification or efficacy result exists. The 278.07-second attempt costs $0.400930 at the recorded $2.50/$15 per million input/output prices; its raw wire response and reconciliation are retained. The high-effort setting was not expanded; the separate low-effort treatment is reported next. The provider documents these settings and prices in the official model reference.

V6 remains the historical original experimental release, with the full V7.1 result alongside it. V7.1 improves warning coverage but lowers supported lexical retention and does not reduce the eight unflagged contradictions. Neither candidate is certified for unattended portfolio decisions. This selection is transparent and does not hide the newer negative result or treat V6 as the best-performing model.

9.2 Same-input reasoning treatment and evaluator corrections

The follow-up made 24 fresh reviews, eight each for NVIDIA, AMD and Intel, with `gpt-5.4-2026-03-05`, low reasoning effort, a 20,000 completion-token ceiling and omitted temperature. It held writer drafts, original prompts and the semantic-review instructions fixed. These calls cost $3.0613925, used 287,867 tokens and took 2,062.10 seconds of model-call time. V8.2 revalidated captured responses offline; this is not another independent generation. The failed high-effort pilot cost $0.400930 and the successful low-effort pilot $0.138385; both remain in the cumulative ledger.

A second audit found that the evaluator sometimes added a time window, comparator or universal quantifier absent from the source sentence. For example, a sentence describing a table that extends into forecasts does not necessarily claim that a displayed CAGR was calculated across that full table. Such under-specified assertions are Unsupported, not automatically Contradicted or Supported. The correction reviewed both before and after, including withdrawing several unsupported Supported labels. An AMD baseline assertion was split into a supported observed decline and an unsupported assertion that the level remained elevated. The saved mapping changes 31 original atoms into 32; 120 original source/audit files retain their hashes. No report text, evidence prompt or model response changes in this measurement correction.

The corrected baseline is 286 Supported, 1,162 Unsupported and 31 Contradicted among 1,479 claims. Corrected V8.2 is 530 Supported, 1,420 Unsupported and 8 Contradicted among 1,958. Its warning recall is 1,416/1,428 (99.16%); seven contradictions and five unsupported assertions have no warning. The previously reported 16 contradictions and 12 unflagged contradictions are preserved historical evaluator results, superseded for this comparison. Changing labels is not a verifier performance gain.

9.3 Narrow deterministic semantic checks within Option A

The verifier now has two narrow deterministic release checks. V9 reads the subject's supplied Revenue row and fiscal A/E header, verifies that the displayed CAGR rounds to the actual-endpoint calculation, and explicitly names both actual and full-table horizons. The calculation is `(ending / starting) ** (1 / year_difference) - 1`; year differences, rather than column counts, determine elapsed intervals. Forecast endpoints remain labeled as model forecasts. For NVIDIA, the historical cell is 68.3% while the 2022A-2029E endpoint calculation is 37.49%; AMD's extended result is 13.59% and Intel's is -3.59%. The twelve matching original sentences are preserved alongside each clarification. Ambiguous originals are not retrospectively declared false.

V9.1 also handles one narrowly matched EV/EBITDA range sentence. It replaces an unspecified recent-years window with an explicitly dated window from the supplied actual rows and calculates its minimum and maximum. The rule reads no audit label, hard-coded ticker result or later information. Missing, inconsistent or ambiguous inputs cause abstention. Original model classifications, numeric bindings, release status and revised prose remain separately saved; these checks do not alter paid instructions or captured responses.

Version, under corrected evaluation	Supported / claims	Grounding	Contradicted	Numeric errors / verifiable
Matched baseline	286 / 1479	19.34%	31	25 / 172
Reasoning V8.2, unchanged prose	530 / 1958	27.07%	8	8 / 430
V9: explicit CAGR horizons	566 / 1982	28.56%	1	1 / 459
V9.1: explicit actual multiple range	570 / 1985	28.72%	0	0 / 461

The final version retains 1985 substantive atomic assertions, 134.21% of the baseline count, including 1415 Unsupported and 0 Contradicted. Numeric claims include 51 that remain unverifiable. A low observed contradiction count is not proof of factual correctness: unknown claims remain unknown, the evaluated cohort informed development, and there is no held-out test. No general semantic-verification guarantee is claimed.

Independent warning recall is 100.00%, precision 89.90%; 0 unsupported/contradicted assertions remain unflagged, including 0 contradictions. Supported lexical-key retention is 9.68%, while source-evidence-span retention is 81.82%. These disagree because rephrasing, expanded endpoint statements and decomposition affect the proxies. Neither is unique-fact recall; claim-count growth does not prove information preservation. Warned claims remain in all substantive denominators.

All eight final sections of each company are audited against their original writer prompts. An identical source unit can inherit its corrected prior label only when the section, text and evidence hashes match. Changed units receive independent decomposition and calculations; the CAGR evaluator uses a separate Decimal logarithm/exponential implementation, not the verifier function. This is task-agent-assisted evaluation, not human validation; its exact task-agent model usage is unavailable in the direct-API ledger. The audit remains fallible.

V9 and V9.1 reuse the same 24 actual paid responses with model-call methods and network connections disabled. V9.1 adds zero API tokens and $0 new model cost; verification plus rendering takes 7.05 seconds across the three saved derivations. The $3.0613925 original-review cost is allocated to this treatment for comparison, not billed again. Separate upstream compatibility and Option A patches apply to the pinned commit; the final Option A patch SHA is `c7d6fab68e32d49f2237cfb201ca8fb25422f823a7b6f09ac4d428e0d046d889`.

The explicit audit correction, V9 and V9.1 scorecards are separate. Original V6/V7.1 and V8.2 prose and labels remain available as experiment history. The final three-company reports are linked from the README; five-company coverage and the original stability experiment are unchanged. Additional safeguards improve traceability within Option A, not a second research option.

10. Cost, runtime and reproducibility

The direct API ledger contains 347 settled calls, 1,303,571 input tokens and 822,575 output tokens, totaling **$10.04507** of the authorized $20, with zero unresolved reservations. Sum of measured model-call time is 5023.18 seconds. This includes preflight, abandoned verifier/judge attempts, repairs and all nine baseline generations. Prices are the saved $0.40/$1.60 per million input/output for mini , $2/$8 for GPT-4.1 and $2.50/$15 for GPT-5.4; cached tokens are conservatively charged full price. It is a reproducible list-price upper estimate, not an invoice. Codex-assisted authoring/adjudication usage is not available in that ledger and is not asserted to be free. No data subscription upgrade was purchased.

Original V6 per-company stage accounting (the later reasoning-treatment allocation is stated in Section 9.3):

Company	Writer tokens in / out	Writer seconds / USD	Allocated verifier seconds / USD	Baseline / final render seconds
NVDA	16,359 / 4,152	31.31 / 0.01319	121.39 / 0.26320	1.78 / 1.73
AMD	16,199 / 4,063	28.11 / 0.01298	96.71 / 0.25315	1.68 / 1.67
INTC	16,063 / 3,976	28.28 / 0.01279	92.52 / 0.24714	1.60 / 1.62
AAPL	15,967 / 3,524	25.30 / 0.01203	90.62 / 0.23053	1.60 / 1.62
GOOGL	16,119 / 3,553	25.06 / 0.01213	81.24 / 0.22559	1.65 / 1.68
AVGO	16,535 / 3,980	28.38 / 0.01298	92.27 / 0.24575	1.51 / 1.51
QCOM	16,319 / 4,073	28.22 / 0.01304	106.33 / 0.29600	1.58 / 1.49

These stages were measured separately, so their sum is a staged reproduction estimate rather than an observed uninterrupted end-to-end service latency. Snapshot analysis times are separately recorded in `report_stage_runtime.csv`. NVIDIA replicates 2 and 3 used 20,261 and 19,894 total tokens, $0.01279 and $0.01220, and 27.21 and 24.38 seconds. For the original V6 paired comparison, the 24 allocated verifier reviews total 170,352 tokens, $0.763494 and 310.62 seconds. V6's incremental paid cost for these three inherited reviews is zero; this does not erase their original paid cost or the global development overhead. New AVGO/QCOM verifier costs are $0.245748 and $0.296004, including a QCOM repair.

The original V6 patches apply to a clean copy of the pinned upstream, with 108 tests and 6 subtests passing. The clean patched implementation then reproduces all seven data pipelines and 14 saved baseline/final reports with sockets disabled (`reproduction/clean7_v3/` execution summarized in saved evidence). Nine relocated independent notebooks execute without upstream imports, pandas or a network, with all original valuation CSV hashes unchanged. A replay defect in derived snapshots was discovered and fixed by resolving and hash-checking the original sibling HTTP capture; failed runs remain logged. No replay is counted as a new generation.

11. What a portfolio manager can use

Accession-linked historical observations, the twenty reconciled data rows, inspectable scenario equations and source/error maps are useful for analyst screening and quality control. A reviewer can reproduce a selected input, locate a discrepant earnings definition and see how a stated assumption moves a target. The warnings offer a measurable triage aid, with substantial false positives and missed errors.

Unattended recommendations, position sizes and fair-value targets remain unreliable. Forecast EBITDA changes definition, R&D is omitted, the peer projection uses incompatible growth mechanics and dates, debt and discounting are crude, subject multiples may be fallbacks, and news is narrow. Machine narratives overreach the supplied evidence. A precise matching calculation and an attractive HTML report do not resolve those weaknesses. Human analysts must validate decisions and material claims; this experiment provides no human certification.

The complete submission dependencies were also copied to a new directory and replayed with Python access to the original submission denied, network and paid calls disabled, and an execution clock forced to 2099. Exact HTML, section text, chart assets, market inputs and valuation hashes are checked against saved originals. The original report date is preserved while new execution time is recorded separately. This closes the earlier

output-directory-only replay gap; the isolation is a tested Python guard, not an operating-system security sandbox.

12. Submission, sources and repeatability

The course submission is the new `Assignment1_ZeyuGu_zg2556` folder, with no nested upstream clone or credentials. `README.md` gives exact clean-install, offline replay, fresh-generation and audit commands. `coverage_universe.csv`, `snapshots/`, `runs/`, `audits/`, `patches/` and the artifact inventory form the evidence chain. Immutable raw inputs and model responses are preserved, including failures; reproducible summary metrics come from saved CSVs and manifests. New online retrieval requires new timestamps/run IDs and may fail or differ as service permissions and market data change. Public-source acquisition scripts record real response bytes; rerunning the frozen adapter itself is offline.

The complete analytical Medium article is a separate discussion of these experiments rather than pasted stock research. Its public URL and the course master PR will be added only after actual publication/submission. Publication is deferred at the user's request, so this document does not declare the entire assignment formally submitted. A strict requirement of zero errors in generated narrative also remains unmet, transparently quantified above. Visual/asset QA and credential scanning have separate manifests and do not substitute for semantic correctness.

Result figures

The following figures are generated from saved experiment results. These are scenario analyses and measured audits; no probability distribution or unperformed result is implied.

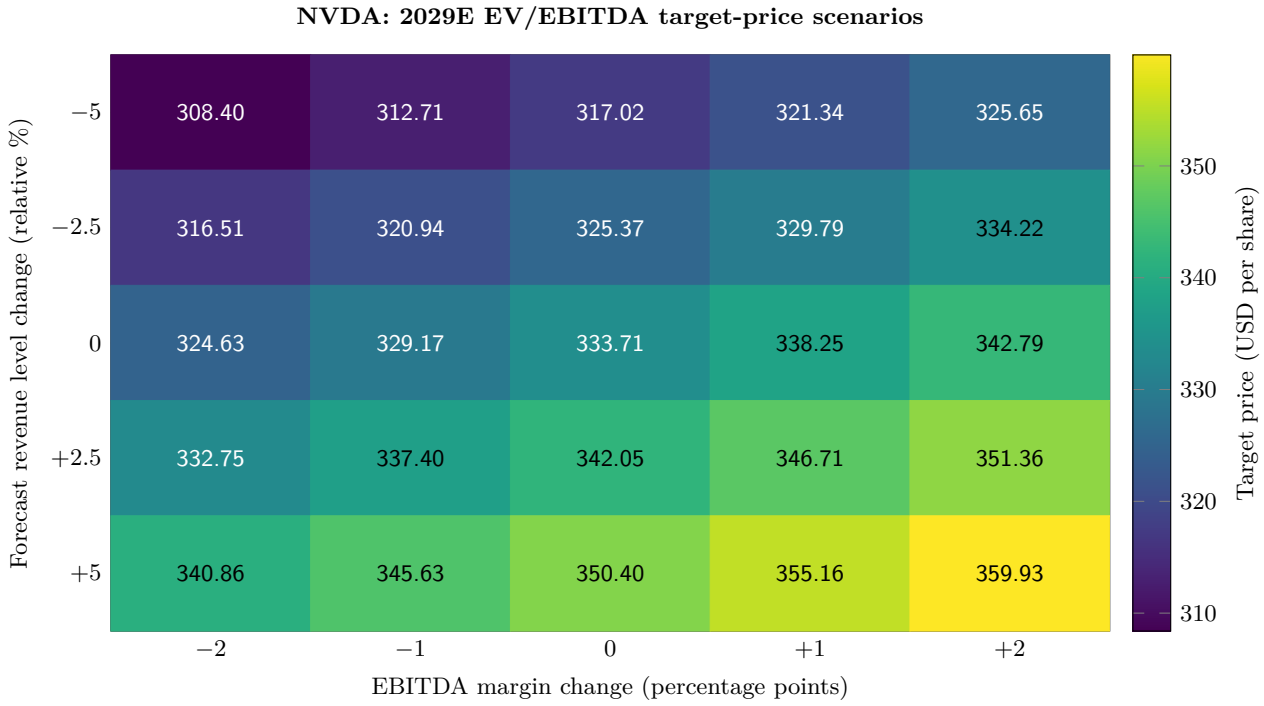


Figure 2: NVDA target-price sensitivity from the saved 2029E EV/EBITDA scenarios. Revenue shocks change the forecast level; margin shocks are percentage points. The multiple and shares are fixed; net debt remains 10% of each scenario's enterprise value. These are conditional scenarios, not confidence intervals.

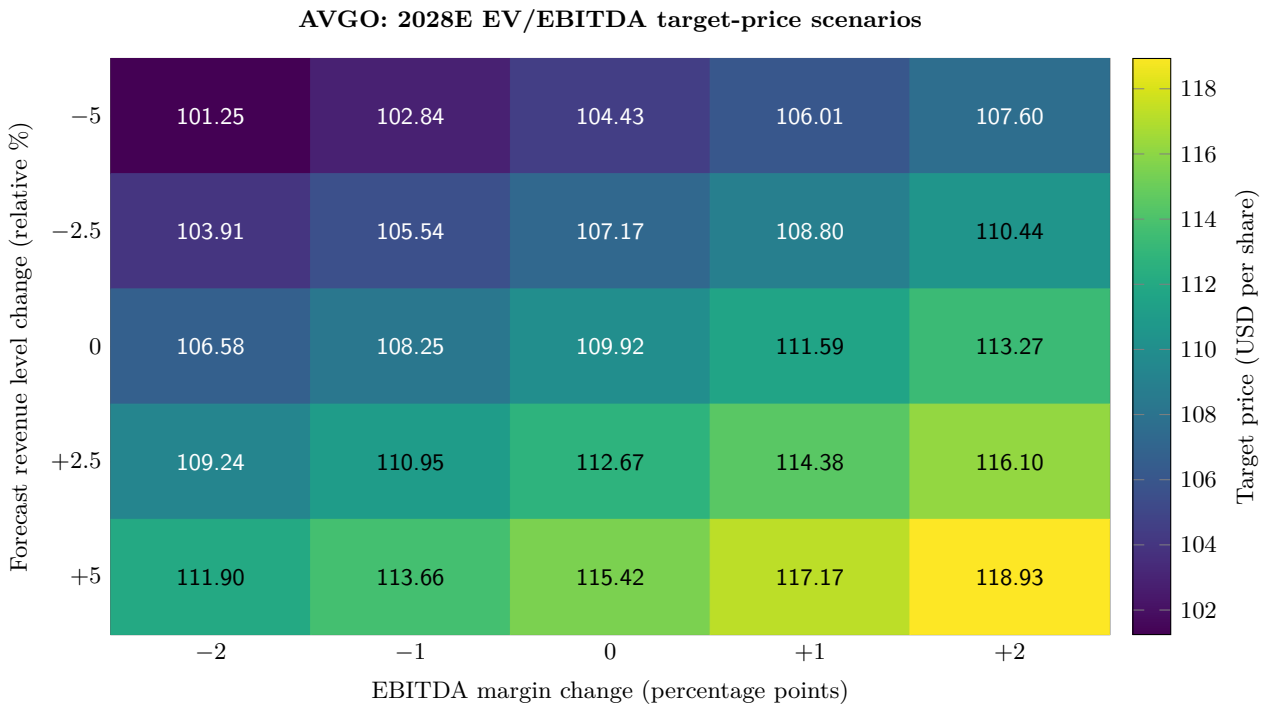


Figure 3: AVGO target-price sensitivity from the saved 2028E EV/EBITDA scenarios. Revenue shocks change the forecast level; margin shocks are percentage points. The multiple and shares are fixed; net debt remains 10% of each scenario's enterprise value. These are conditional scenarios, not confidence intervals.